

ANSH PARASHAR

Full-Stack Engineer • AI/ML Engineering

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SUMMARY

Final-year B.Tech Computer Science (AI) student who builds full-stack products end to end, then trains, evaluates, and wires in whatever machine learning the product needs — rather than treating the model as a separate deliverable. Shipped projects spanning MERN applications, an AI code-analysis tool, a RAG pipeline, and applied ML systems, most deployed and publicly demoable; currently building an enterprise AI-agent platform focused on reliability and observability.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, SQL, C/C++
- **Full-Stack & Web:** React (Vite), Node.js, Express.js, MongoDB / MongoDB Atlas, REST APIs, Socket.IO, JWT auth, Tailwind CSS, HTML5/CSS3
- **AI & ML Engineering:** Retrieval-Augmented Generation (RAG) pipelines, LLM integration (Llama 3.1 via Groq, Gemini), tree-sitter/AST parsing, Random Forest, XGBoost, CNN-LSTM, leakage-safe cross-validation
- **Data & Algorithms:** Dijkstra's algorithm, Yen's k-shortest-paths, exploratory data analysis, Pandas, scikit-learn
- **Tools & Platforms:** Git/GitHub, ChromaDB, Streamlit, Google Colab, VS Code, Netlify, Render, automated testing & CI

PROJECTS

CodeViz AI — Full-stack + AI

React 19, Vite, ReactFlow, Express, TypeScript, tree-sitter, Gemini API

- Built an AI-powered code visualizer that turns a GitHub repo into an interactive multi-file AST dependency graph, with per-node explanations in technical or plain-language mode.
- Migrated parsing from regex to tree-sitter across Python, Java, JavaScript, and TypeScript; shipped with 123 passing tests and zero regressions.
- Deployed live — codeviz-ai.netlify.app • [Source](#)

Logistics Route Optimizer — Full-stack + Algorithms

React (Vite), Tailwind, react-leaflet, Express, MongoDB Atlas, OSRM

- Built a MERN routing app that plots the fastest path between multiple stops using real road geometry via the OSRM API, not straight-line estimates.
- Implemented Dijkstra's shortest-path algorithm and Yen's k-shortest-paths so users can compare the top route against ranked alternatives.
- Deployed live — logistics-route-optimizer.netlify.app • [Source](#)

Interactive Job Portal — Full-stack (team project)

React (Vite), Node.js, Express, MongoDB

- Built the React (Vite) frontend for a job-matching platform connecting recruiters and applicants, as one of a 4-person final-year team.
- Implemented candidate profile views and client-side filtering by role, location, and requirements; backend/API integration is ongoing with the team.
- Source — github.com/Ansh-san/Job_Portal

RAG PDF Chatbot — AI / LLM Engineering

Python, Streamlit, PyMuPDF, sentence-transformers, ChromaDB, Llama 3.1 (Groq)

- Built a retrieval-augmented chatbot that answers questions about any uploaded PDF using only the document's own text, avoiding hallucinated answers.
- Implemented the full pipeline from scratch — chunking, MiniLM embeddings, ChromaDB indexing, and top-k retrieval feeding Llama 3.1 8B via the Groq API.
- Deployed on Streamlit Community Cloud • [Source](#)

Temperature Prediction Engine — Data & Analytics

Python, Pandas, scikit-learn, Streamlit

- Built a hybrid forecasting model — Random Forest for the historical range, per-country linear trend extrapolation beyond it — across 196 countries, 1950–2024.
- Achieved test R^2 of 0.73 and MAE of 3.15°C; shipped as an interactive Streamlit app with prediction, world-globe, and country-trend views.
- Deployed on Streamlit Community Cloud · [Source](#)

Exoplanet Detection — Kepler — Data & Machine Learning

Python, Pandas, scikit-learn, XGBoost, TensorFlow

- Trained an ensemble of Random Forest, XGBoost, and CNN-LSTM models to detect exoplanet transit signals in NASA Kepler light-curve data.
- Evaluated with stratified 5-fold cross-validation and precision-recall curves after identifying and correcting a data-leakage issue in the original pipeline.
- Source — github.com/Ansh-san/Exoplanet-Detection-Kepler

Also shipped a real-time MERN chat app (Socket.IO, JWT auth) — solo project, not listed above for space.

CURRENTLY BUILDING

OpsPilot — AI WorkOps Platform (in progress)

Planned: React + TypeScript, FastAPI, PostgreSQL + pgvector, LangGraph, Docker, AWS

- Designing an enterprise AI-agent platform that investigates IT/business incidents end-to-end — classification, RAG-based retrieval over internal docs, tool-calling investigation, and a recommended fix — with a human-approval step before any action executes.
- Core focus is reliability over demo polish: an audit trail for every agent decision, a risk-based policy engine (auto-execute / approve / block), and an evaluation layer tracking groundedness, tool-call success, and cost per request.
- Currently building the MVP (ticket → classification → RAG → recommendation) before layering in agents, the approval system, and observability in later stages.

VIRTUAL INTERNSHIPS & PROFESSIONAL DEVELOPMENT

- **Google AI-ML Virtual Intern** — EduSkills, in collaboration with Google for Developers. 10-week program in applied ML and TensorFlow; 88% completion across technical modules.
- **ServiceNow Virtual Intern** — SmartBridge, in collaboration with AICTE. Configured workflow automation using the Automated Test Framework (ATF); completed CSA certification prep.
- **Data Analytics Job Simulation** — Deloitte (via Forage). Analyzed business datasets and prepared stakeholder-facing findings.

EDUCATION

Meerut Institute of Engineering & Technology (MIET)

B.Tech, Computer Science & Engineering (Artificial Intelligence) · CGPA 7.1 · Expected 2027

CERTIFICATIONS

- AI Engineer — Agentic & Core Tracks (LLM Engineering, RAG, QLoRA, Agents, MCP), Udemy
- Generative AI Essentials, TCS iON
- CyberOps Associate, Cisco Networking Academy
- MERN Stack Development, Anudip Foundation
- Java Full Stack Development, Coursera